Unit- 04: Network Models

CONTENTS

Objectives

After studying this unit, you will be able to:

- Discuss concept of process network software and the significance of layering the communication process and related design issues for the layers
- Describe different technologies involved in defining the network hardware
- Explain what are the reference models for computer networks and how they are related with the OSI reference model

Introduction

A network is a combination of hardware and software that sends data from one location to another. The hardware consists of the physical equipment that carries signals from one point of the network to another. The software consists of instruction sets that make possible the services that we expect from a network.We can compare the task of networking to the task of solving a mathematics problem with a computer. The fundamental job of solving the problem with a computer is done by computer hardware. However, this is a very tedious task if only hardware is involved. We would need switches for every memory location to store and manipulate data. The task is much easier if software is available. At the highest level, a program can direct the problem-solving process; the details of how this is done by the actual hardware can be left to the layers of software that are called by the higher levels. Compare this to a service provided by a computer network. For example, the task of sending an e-mail from one point in the world to another can be broken into several tasks, each performed by a separate software package. Each software package uses the services of another software package. At the lowest layer, a signal, or a set of signals, is sent from the source computer to the destination computer. In this chapter, we give a general idea of the layers of a network and discuss the functions of each.

4.1 Layered Tasks

We use the concept of layers in our daily life. As an example, let us consider two friends who communicate through postal maiL The process of sending a letter to a friend would be complex if there were no services available from the post office. *Figure 1*shows the steps in this task.

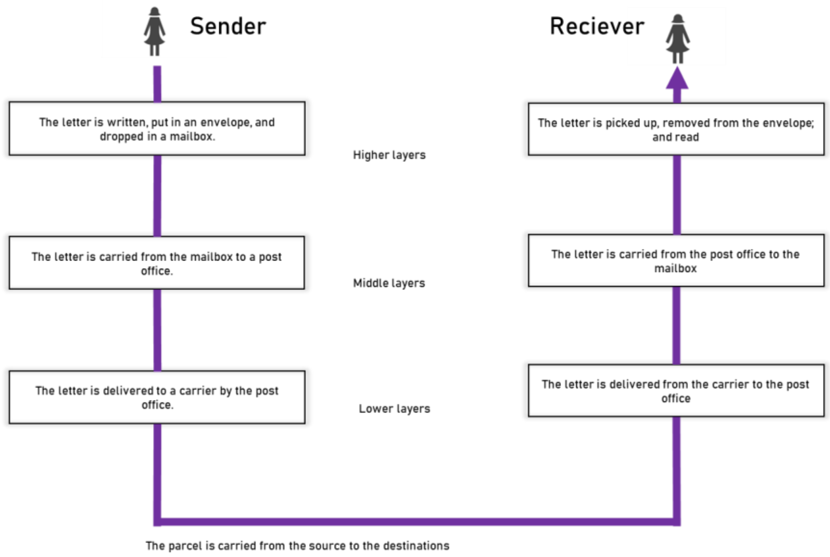

Figure 1 Tasks Involve in Sending a Letter

Sender, Receiver, and Carrier

In *Figure 1* we have a sender, a receiver, and a carrier that transports the letter. There is a hierarchy of tasks.

At the Sender Site

Let us first describe, in order, the activities that take place at the sender site.

o Higher layer. The sender writes the letter, inserts the letter in an envelope, writes the sender and receiver addresses, and drops the letter in a mailbox.

o Middle layer. The letter is picked up by a letter carrier and delivered to the post office.

o Lower layer. The letter is sorted at the post office; a carrier transports the letter.

On the Way

The letter is then on its way to the recipient. On the way to the recipient's local post office, the letter may actually go through a central office. In addition, it may be transported by truck, train, airplane, boat, or a combination of these.

At the Receiver Site

- Lower layer. The carrier transports the letter to the post office.
- Middle layer. The letter is sorted and delivered to the recipient's mailbox.
- Higher layer. The receiver picks up the letter, opens the envelope, and reads it.

Hierarchy

According to our analysis, there are three different activities at the sender site and another three activities at the receiver site. The task of transporting the letter between the sender and the receiver is done by the carrier. Something that is not obvious immediately is that the tasks must be done in the order given in the hierarchy. At the sender site, the letter must be written and dropped in the mailbox before being picked up by the letter carrier and delivered to the post office. At the receiver site, the letter must be dropped in the recipient mailbox before being picked up and read by the recipient.

Services

Each layer at the sending site uses the services of the layer immediately below it. The sender at the higher layer uses the services of the middle layer. The middle layer uses the services of the lower layer. The lower layer uses the services of the carrier. The layered model that dominated data communications and networking literature before 1990 was the Open Systems Interconnection

(OSI) model. Everyone believed that the OSI model would become the ultimate standard for data communications, but this did not happen. The TCPIIP protocol suite became the dominant commercial architecture because it was used and tested extensively in the Internet; the OSI model was never fully implemented.

4.2 The OSI Model

Established in 1947, the International Standards Organization (ISO) is a multinational body dedicated to worldwide agreement on international standards. An ISO standard that covers all aspects of network communications is the Open Systems Interconnection model. It was first introduced in the late 1970s. An open system is a set of protocols that allows any two different systems to communicate regardless of their underlying architecture. The purpose of the OSI model is to show how to facilitate communication between different systems without requiring changes to the logic of the underlying hardware and software. The OSI model is not a protocol; it is a model for understanding and designing a network architecture that is flexible, robust, and interoperable.

ISO is the organization

OSI is the model

The OSI model is a layered framework for the design of network systems that allows communication between all types of computer systems. It consists of seven separate but related layers, each of which defines a part of the process of moving information across a network (seeFigure 2). An understanding of the fundamentals of the OSI model provides a solid basis for exploring data communications.

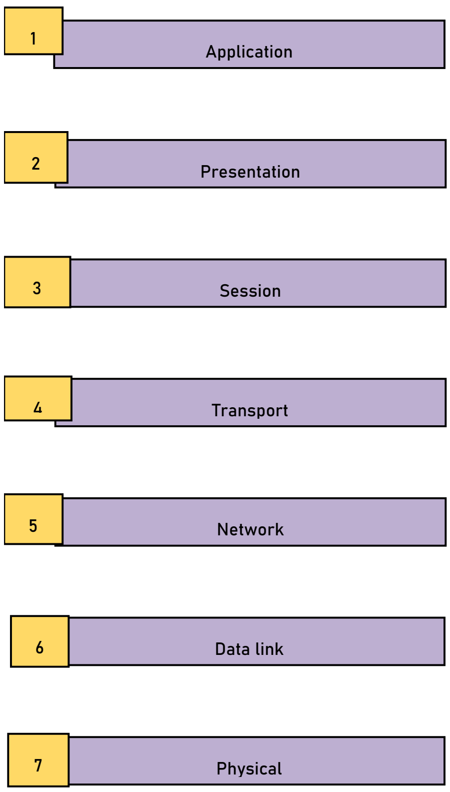

Figure 2 Seven Layers of OSI Model

Layered Architecture

The OSI model is composed of seven ordered layers: physical (layer 1), data link (layer 2), network (layer 3), transport (layer 4), session (layer 5), presentation (layer 6), and application (layer 7). Figure 2.3 shows the layers involved when a message is sent from device A to device B. As the message travels from A to B, it may pass through many intermediate nodes. These intermediate nodes usually involve only the first three layers of the OSI model.

In developing the model, the designers distilled the process of transmitting data to its most fundamental elements. They identified which networking functions had related uses and collected those functions into discrete groups that became the layers. Each layer defines a family of functions distinct from those of the other layers. By defining and localizing functionality in this fashion, the designers created an architecture that is both comprehensive and flexible. Most importantly, the OSI model allows complete interoperability between otherwise incompatible systems. Within a single machine, each layer calls upon the services of the layer just below it. Layer 3, for example, uses the services provided by layer 2 and provides services for layer 4. Between machines, layer x on one machine communicates with layer x on another machine. This communication is governed by an agreed-upon series of rules and conventions called protocols. The processes on each machine thatcommunicate ata given layer are called peer-to-peer processes. Communication between machines istherefore a peer-to-peer process using the protocols appropriate to a given layer.

Peer-to-Peer Processes

At the physical layer, communication is direct: In Figure 2.3, device A sends a stream of bits to device B (through intermediate nodes). At the higher layers, however, communication must move down through the layers on device A, over to device B, and thenback up through the layers.

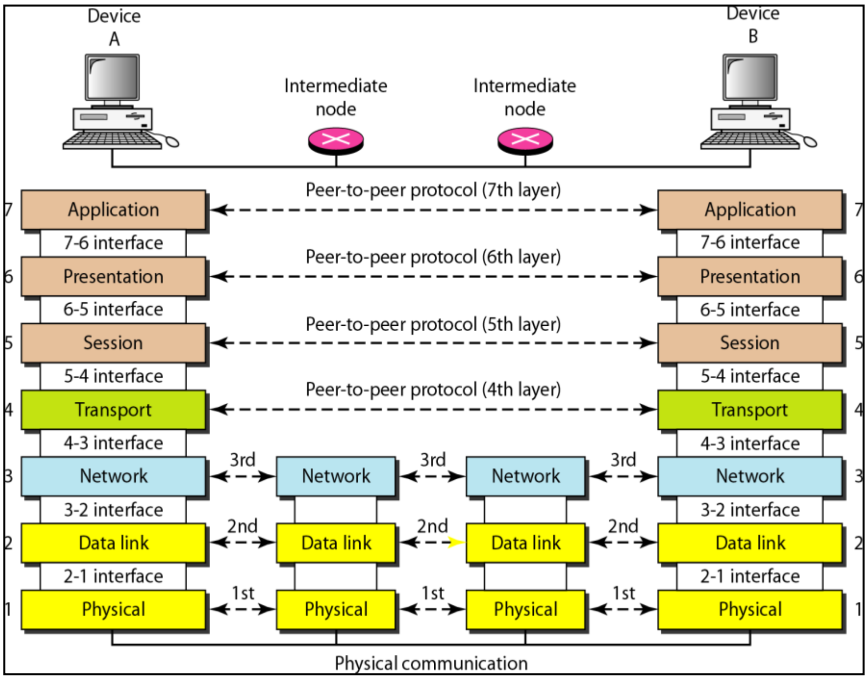

Figure 3 The interaction between layers in OSI Model

Each layer in the sending device adds its own information to the message it receives from the layer just above it and passes the whole package to the layer just below it. At layer I the entire package is converted to a form that can be transmitted to the receiving device. At the receiving machine, the message is unwrapped layer by layer, with each process receiving and removing the data meant for it. For example, layer 2 removes the data meant for it, then passes the rest to layer 3. Layer 3 then removes the data meant for it and passes the rest to layer 4, and so on.

Interfaces Between Layers

The passing of the data and network information down through the layers of the sending device and back up through the layers of the receiving device is made possible by an interface between each pair of adjacent layers. Each interface defines the information and services a layer must provide for the layer above it. Well-defined interfaces and layer functions provide modularity to a network. As long as a layer provides the expected services to the layer above it, the specific implementation of its functions can be modified or replaced without requiring changes to the surrounding layers.

Organization of the Layers

The seven layers can be thought of as belonging to three subgroups. Layers I, 2, and 3-physical, data link, and network-are the network support layers; they deal with the physical aspects of moving data from one device to another (such as electrical specifications, physical connections, physical addressing, and transport timing and reliability). Layers 5, 6, and 7-session, presentation, and application-can be thought of as the user support layers; they allow interoperability among unrelated software systems. Layer 4, the transport layer, links the two subgroups and ensures that what the lower layers have transmitted is in a form that the upper layers can use. The upper OSI layers are almost always implemented in software; lower layers are a combination of hardware and software, except for the physical layer, which is mostly hardware. In Figure 4, which gives an overall view of the OSI layers, D7 means the data unit at layer 7, D6 means the data unit at layer 6, and so on. The process starts at layer 7 (theapplication layer), then moves from layer to layer in descending, sequential order. At each layer, a header, or possibly a trailer, can be added to the data unit. Commonly, the trailer is added only at layer 2. When the formatted data unit passes through the physical layer (layer 1), it is changed into an electromagnetic signal and transported along a physical link.

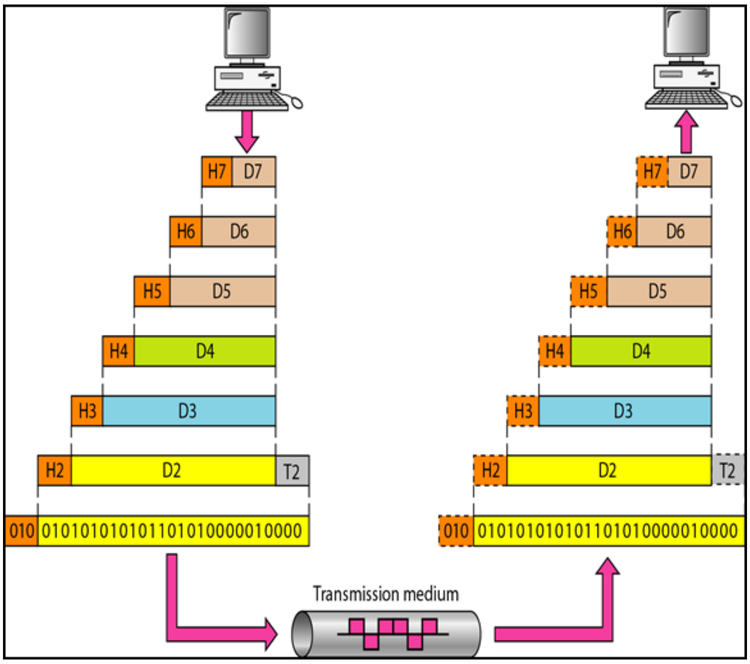

Figure 4 An exchange using the OSI model

Upon reaching its destination, the signal passes into layer 1 and is transformed back into digital form. The data units then move back up through the OSI layers. As each block of data reaches the next higher layer, the headers and trailers attached to it at the corresponding sending layer are removed, and actions appropriate to that layer are taken. By the time it reaches layer 7, the message is again in a form appropriate to the application and is made available to the recipient.

Encapsulation

Figure 2.3 reveals another aspect of data communication in the OSI model: encapsulation. A packet (header and data) at level 7 is encapsulated in a packet at level 6. The whole packet at level 6 is encapsulated in a packet at level 5, and so on. In other words, the data portion of a packet at level N - 1 carries the whole packet (data and header and maybe trailer) from level N. The concept is called encapsulation; level N - 1 is not aware of which part of the encapsulated packet is data and which part is the header or trailer. For level N - 1, the whole packet coming from level N is treated as one integral unit.

4.3 Layers in the OSI Model

In this section we briefly describe the functions of each layer in the OSI model.

Physical Layer

The physical layer coordinates the functions required to carry a bit stream over a physical medium. It deals with the mechanical and electrical specifications of the interface and transmission medium. It also defines the procedures and functions that physical devices and interfaces have to perform for transmission to Occur. Figure 5 shows the position of the physical layer with respect to the transmission medium and the data link layer.

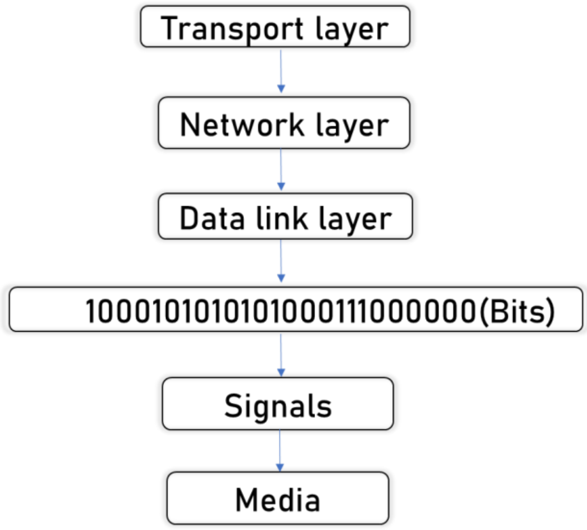

Figure 5 Physical Layer

The physical layer is also concerned with the following:

Physical characteristics of interfaces and medium. The physical layer defines the characteristics of the interface between the devices and the transmission medium. It also defines the type of transmission medium.

Representation of bits. The physical layer data consists of a stream of bits (sequence of Os or 1s) with no interpretation. To be transmitted, bits must be encoded into signals--electrical or optical. The physical layer defines the type of encoding (how Os and I s are changed to signals).

Data rate. The transmission rate-the number of bits sent each second-is also defined by the physical layer. In other words, the physical layer defines the duration of a bit, which is how long it lasts.

Synchronization of bits. The sender and receiver not only must use the same bit rate but also must be synchronized at the bit level. In other words, the sender and the receiver clocks must be synchronized.

Line configuration. The physical layer is concerned with the connection of devices to the media. In a point-to-point configuration, two devices are connected through a dedicated link. In a multipoint configuration, a link is shared among several devices.

Physical topology. The physical topology defines how devices are connected to make a network. Devices can be connected by using a mesh topology (every device is connected to every other device), a star topology (devices are connected through a central device), a ring topology (each device is connected to the next, forming a ring), a bus topology (every device is on a common link), or a hybrid topology (this is a combination of two or more topologies).

Transmission mode. The physical layer also defines the direction of transmission between two devices: simplex, half-duplex, or full-duplex. In simplex mode, only one device can send; the other can only receive. The simplex mode is a one-way communication. In the half-duplex mode, two devices can send and receive, but not at the same time. In a full-duplex (or simply duplex) mode, two devices can send and receive at the same time.

Data Link Layer

The data link layer transforms the physical layer, a raw transmission facility, to a reliable link. It makes the physical layer appear error-free to the upper layer (network layer). Figure 6shows the relationship of the data link layer to the network and physical layers.

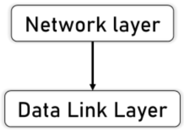

Figure 6 Data Link Layer

Other responsibilities of the data link layer include the following:

- ***Framing***. The data link layer divides the stream of bits received from the network layer into manageable data units called frames.
- ***Physical addressing***. If frames are to be distributed to different systems on the network, the data link layer adds a header to the frame to define the sender and/or receiver of the frame. If the frame is intended for a system outside the sender's network, the receiver address is the address of the device that connects the network to the next one.
- ***Flow control***. If the rate at which the data are absorbed by the receiver is less than the rate at which data are produced in the sender, the data link layer imposes a flow control mechanism to avoid overwhelming the receiver.
- ***Error control***. The data link layer adds reliability to the physical layer by adding mechanisms to detect and retransmit damaged or lost frames. It also uses a mechanism to recognize duplicate frames. Error control is normally achieved through a trailer added to the end of the frame.
- ***Access control***. When two or more devices are connected to the same link, data link layer protocols are necessary to determine which device has control over the link at any given time.

Addressing: It can be of two types as shown in .Networks operate in exactly the same way. The physical address of a computer on a LAN, fixed in the network interface card by the manufacturer, works like physical knowledge of where a house is. Messages sent on the same LAN segment, corresponding to a house's neighborhood, can get to a specific physical LAN address. If you want to send a message across the country or around the world, though, you need the equivalent of a country, state, city, and street address. In network terms, the address that you need is called a logical address. The key difference between physical addresses and logical addresses is that, although physical addresses are scattered randomly around the world, logical addresses follow a pattern determined by network administrators and stored in routing tables. Routing tables (used by routers) are the equivalent of street maps, guiding messages to their destination.

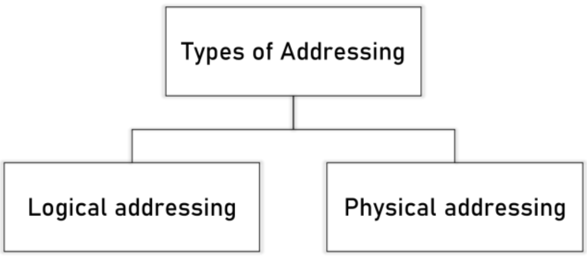

Figure 7 Types of Addressing

Note The data link layer is responsible for moving frames from one hop to the next.

Network Layer

The network layer is responsible for the source-to-destination delivery of a packet, possibly across multiple networks (links). Whereas the data link layer oversees the delivery of the packet between two systems on the same network (links), the network layer ensures that each packet gets from its point of origin to its final destination. If two systems are connected to the same link, there is usually no need for a network layer. However, if the two systems are attached to different networks (links) with connecting devices between the networks (links), there is often a need for the network layer to accomplish source-to-destination delivery. Figure 8 shows the relationship of the network layer to transport layers.

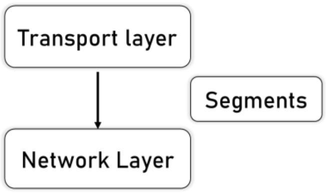

Figure 8 Network Layer

Other responsibilities of the network layer include the following:

Logical addressing. The physical addressing implemented by the data link layer handles the addressing problem locally. If a packet passes the network boundary, we need another addressing system to help distinguish the source and destination systems. The network layer adds a header to the packet coming from the upper layer that, among other things, includes the logical addresses of the sender and receiver.

Routing. When independent networks or links are connected to create *internetworks* (network of networks) or a large network, the connecting devices (called *routers*or *switches)* route or switch the packets to their final destination. One of the functions of the network layer is to provide this mechanism. Figure 9 illustrates end-to-end delivery by the network layer.

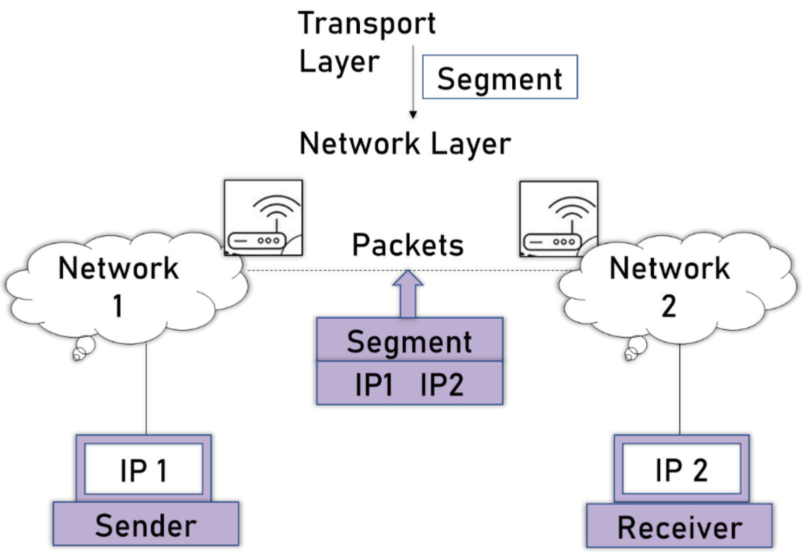

Figure 9 Source to destination delivery

Transport Layer

The transport layer is responsible for process-to-process delivery of the entire message. A process is an application program running on a host. Whereas the network layer oversees source-to-destination delivery of individual packets, it does not recognize any relationship between those packets. Ittreats each one independently, as though each piece belonged to a separate message, whether or not it does. The transport layer, on the other hand, ensures that the whole message arrives intact and in order, overseeing both error control and flow control at the source-to-destination level. Figure 10shows the relationship of the transport layer to the network and session layers.

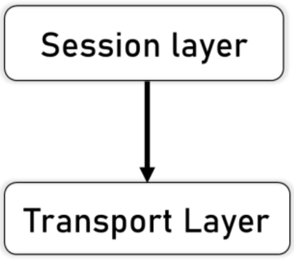

Figure 10 Transport Layer

Other responsibilities of the transport layer include the following:

- ***Service-point addressing***. Computers often run several programs at the same time. For this reason, source-to-destination delivery means delivery not only from one computer to the next but also from a specific process (running program) on one computer to a specific process (running program) on the other. The transport layer header must therefore include a type of address called a *service-point address* (or port address). The network layer gets each packet to the correct computer; the transport layer gets the entire message to the correct process on that computer.
- ***Segmentation and reassembly***. A message is divided into transmittable segments, with each segment containing a sequence number as shown in Figure 11. These numbers enable the transport layer to reassemble the message correctly upon arriving at the destination and to identify and replace packets that were lost in transmission.

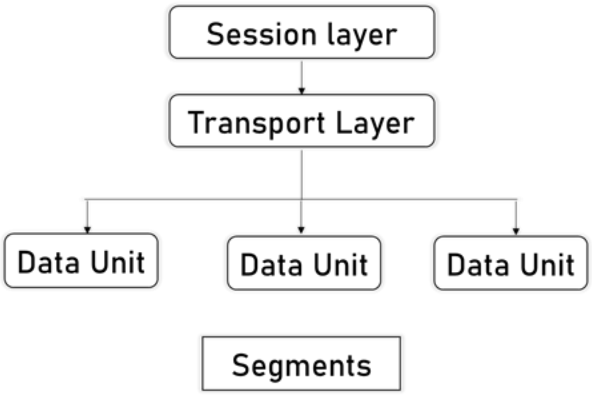

Figure 11 Segmentation

- ***Connection control***. The transport layer can be either connectionless or connection-oriented. A connectionless transport layer treats each segment as an independent packet and delivers it to the transport layer at the destination machine. A connection-oriented transport layer makes a connection with the transport layer at the destination machine first before delivering the packets. After all the data are transferred, the connection is terminated.
- ***Flow control***. Like the data link layer, the transport layer is responsible for flow control. However, flow control at this layer is performed end to end rather than across a single link.
- ***Error control.*** Like the data link layer, the transport layer is responsible for error control. However, error control at this layer is performed process-to-process rather than across a single link. The sending transport layer makes sure that the entire message arrives at the

receiving transport layer without error (damage, loss, or duplication). Error correction is usually achieved through retransmission.

Protocols of Transport Layer

The protocols of transport layer along with difference is explained below *Table 1*.

Table 1 Protocols of Transport Layer

TCP	UDP
Connection Oriented transmission	Connectionless transmission
TCP is slower	UDP is faster
Acknowledgment	No Acknowledgment
Used where full data delivery is must	Does not matter whether we gave received all data of Transport Layer.

Note: The transport layer is responsible for the delivery of a message from one process to another.

Session Layer

The services provided by the first three layers (physical, data link, and network) are not sufficient for some processes. The session layer is the network *dialog controller.* It establishes, maintains, and synchronizes the interaction among communicating systems.

Specific responsibilities of the session layer include the following:

- Dialog control. The session layer allows two systems to enter into a dialog. It allows the communication between two processes to take place in either half duplex (one way at a time) or full-duplex (two ways at a time) mode.
- Authentication and Authorization: Authentication means confirming your own identity, while authorization means granting access to the system. In simple terms, authentication is the process of verifying who you are, while authorization is the process of verifying what you have access to.

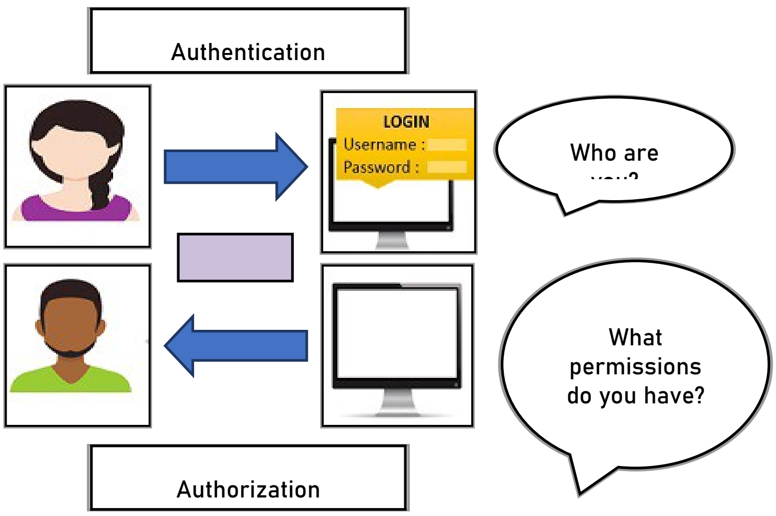

Figure 12 Authentication and Authorization

- Session management
- Synchronization. The session layer allows a process to add checkpoints, or synchronization points, to a stream of data. For example, if a system is sending a file of 2000 pages, it is advisable to insert checkpoints after every 100 pages to ensure that each 100-page unit is received and acknowledged independently. In this case, if a crash happens during the transmission of page 523, the only pages that need to be resent after system recovery are pages 501 to 523. Pages previous to 501 need not be resent. Figure 2.12 illustrates the relationship of the session layer to the transport and presentation layers.

Presentation Layer

The presentation layer performs functions related to the syntax and semantics of the information transmitted that include formatting and displaying of received data by terminals and printers. It is concerned with differences in the data syntax used by communicating applications. This layer is responsible for remedying those differences by resorting to mechanisms that transform the local syntax (specific to the platform in question) to a common one for the purpose of data exchange. For example, it performs encoding of data in a standard agreed upon way to facilitate information exchange among heterogeneous systems using different codes for strings, for example, conversion between ASCII and EBCDIC character codes. It facilitates data compression for reducing the number of bits to be transmitted and encrypts data for privacy and authentication, if necessary.

Translation.The processes (running programs) in two systems are usually exchanging information in the form of character strings, numbers, and so on. The information must be changed to bit streams before being transmitted. Because different computers use different encoding systems, the presentation layer is responsible for interoperability between these different encoding methods. The presentation layer at the sender changes the information from its sender-dependent format into a common format. The presentation layer at the receiving machine changes the common format into its receiver-dependent format.

Encryption. To carry sensitive information, a system must be able to ensure privacy. Encryption means that the sender transforms the original information toanother form and sends the resulting message out over the network. Decryption reverses the original process to transform the message back to its original form.

Compression. Data compression reduces the number of bits contained in the information. Data compression becomes particularly important in the transmission of multimedia such as text, audio, and video.

Application Layer

The application layer provides support services for user and application tasks. It determineshow the user will use the data network. It allows the user to use the network. For example, itprovides network-based services to the end user.Examples of network services are distributed databases, electronic mail, resource sharing, filetransfers, remote file access and network management. This layer defines the nature of the task to be performed.

4.4 TCP/IP

The OSI Model we just looked at is just a reference/logical model. It was designed to describe the functions of the communication system by dividing the communication procedure into smaller and simpler components. But when we talk about the TCP/IP model, it was designed and developed by Department of Defense (DoD) in 1960s and is based on standard protocols. The layers in the TCP/IP protocol suite do not exactly match those in the OSI model. The original TCP/IP protocol suite was defined as having four layers: host-to- network, internet, transport, and application. However, when TCP/IP is compared to OSI, we can say that the TCP/IP protocol suite is made of five layers: physical, data link, network, transport, and application. TCP/IP Model shown inFigure 13.

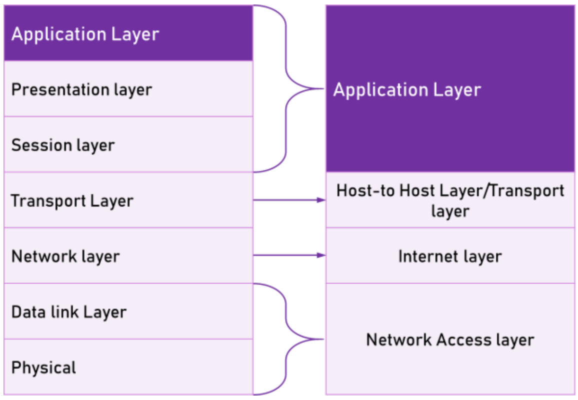

Figure 13 TCP/IP Model (b)

Internet Layer

The packet format and protocol at this layer is called Internet Protocol (IP). IP is a connectionless type service that introduces IP packets into any network. The packets travel independently to the destination. Prior to transmission of data, no logical connection is needed. The TCP/IP Internet layer corresponds to the network layer of the OSI reference model in functionality, as shown in Figure 13.

Transport Layer Notes

The transport layer of TCP/IP model corresponds to the transport layer of the OSI reference model. It is represented by two end-to-end protocols namely, TCP (Transmission Control Protocol) and UDP (User Datagram Protocol). TCP is a reliable connection-oriented protocol and UDP is an unreliable connectionless protocol.

Application Layer

The TCP/IP model was the first of its kind model and therefore did not contain session or presentation layers because of its little use to most of the applications. This layer has all the higher-level protocols, as shown in Figure 13.

Network Access Layer

The layer below the Internet layer is not defined and varies from host and network to network. The TCP/IP model suggests that the host has to connect to the network using some protocol so it can send IP packets over it.

4.5 Similarities and Difference between OSI and TCP/IP

It stands for Transmission Control Protocol/Internet Protocol. The TCP/IP model is a concise version of the OSI model. It contains four layers, unlike seven layers in the OSI model.

Similarities between OSI and TCP/IP

- They share similar architecture.
- They share a common application layer.
- Knowledge by networking professionals.
- Both models assume that packets are switched.

OSI VS TCP/IP

OSI Model	TCP/IP Model
OSI appears to complex because of 7 layers	TCP/IP appears to be a more simpler model and this is mainly due to the fact that it has fewer layers.

Networks are not usually built around the OSI model as it is merely used as a guidance tool.	TCP/IP is considered to be a more credible model- This is mainly due to the fact because TCP/IP protocols are the standards around which the internet was developed therefore it mainly gains creditability due to this reason
The OSI model is bottom to up process of network connection.	TCP/IP is the top to bottom process structure for internet purpose.
OSI defines several more layers of standardized functions	TCP/IP makes no assumptions about what happens above the level of a network session
OSI model is a reference model.	TCP/IP is an implementation of OSI model.
OSI provides layer functioning and also defines functions of all the layers.	TCP/IP model is more based on protocols and protocols are not flexible with other layers.
In OSI model the transport layer guarantees the delivery of packets	In TCP/IP model the transport layer does not guarantees delivery of packets
Follows horizontal approach	Follows vertical approach.
OSI model defines services, interfaces and protocols very clearly and makes clear distinction between them	In TCP/IP it is not clearly separated its services, interfaces and protocols
OSI is a general model	TCP/IP model cannot be used in any other application.
Network layer of OSI model provide both connection oriented and connectionless service.	The Network layer in TCP/IP model provides connectionless service.
It has 7 layers	It has 4 layers
Transport layer guarantees delivery of packets	Transport layer does not guarantees delivery of packets
The protocol are better hidden and can be easily replaced as the technology changes.	It is not easy to replace the protocols
OSI truly is a general model	TCP/IP can not be used for any other

OSI model represents an idea	application TCP/IP network model represents reality in the world

Summary

- To set up a computer network, you'll need three fundamental components: hardware, protocols (software), and applications (useful software). The importance of the idea of layers in networking is also discussed.
- Each two-layer layer serves as an interface for the top layer, allowing each layer to alter with minimal influence on the above levels. This security can be so effective that a programme may be unaware that it is operating on separate hardware.
- There are seven levels in the OSI network paradigm.
- Transmission Control Protocol/Internet Protocol (TCP/IP) is the acronym for Transmission Control Protocol/Internet Protocol. It was created with the goal of defining a set of protocols capable of enabling transparent communications interoperability between computers of any size, independent of the hardware or operating system platforms that support them.
- TCP/IP has grown in popularity over time to become the most widely used protocol today. The public availability of TCP/protocol IP's specifications is one of the reasons for its popularity. TCP/IP may legitimately be called an open system in this regard. TCP/IP is used by most users for file transfers, electronic mail (e-mail), and remote login services.

Keywords

- The Internet protocol suite is a collection of communication protocols that are used on the Internet and other comparable networks.
- Reference Model for Open Systems Interconnection (OSI): The OSI model of data transmission was created by the International Standardization Organization (ISO) in 1984. OSI defines a seven-layer model for defining protocol architectures and functional features, which is utilised by the industry as a frame of reference.
- TCP/IP: Technically, Transmission Control Protocol (TCP) and Internet Protocol (IP) are two separate network protocols. TCP and IP, on the other hand, are so widely used together that TCP/IP has become standard nomenclature to refer to any or both protocols.

Self-Assessment

1. OSI stands for __________

A. open system interconnection
B. operating system interface
C. optical service implementation
D. open service Internet

2. TCP/IP model does not have ______ layer but OSI model have this layer.

A. session layer
B. transport layer
C. application layer

D. network layer

3. Which layer is used to link the network support layers and user support layers?

A. session layer

B. data link layer

C. transport layer

D. network layer

4. TCP/IP model was developed _____ the OSI model.

A. prior to

B. after

C. simultaneous to

D. with no link to

5. Which layer is responsible for process to process delivery in a general network model?

A. network layer

B. transport layer

C. session layer

D. data link layer

6. Which address is used to identify a process on a host by the transport layer?

A. physical address

B. logical address

C. port address

D. specific address

7. Transmission data rate is decided by ____________

A. network layer

B. physical layer

C. data link layer

D. transport layer

8. A layer of the OSI model on one system communicates with the ___ layer of its peer system.

A. above

B. below

C. same

D. None

9. TCP/IP model does not have ______ layer but OSI model have this layer.

A. session layer

B. presentation layer

C. application layer

D. both (a) and (b)

10. Which layer provides the services to user?

A. application layer

B. session layer

C. presentation layer

D. none of the mentioned

11. The number of layers in Internet protocol stack

A. 5

B. 7

C. 6

D. None of the mentioned

12. Transport layer is implemented in

A. End system

B. NIC

C. Ethernet

D. None of the mentioned

Answers for Self Assessment

1. A	2. A	3. C	4. A	5. B
6. C	7. B	8. C	9. D	10. A
11. A	12. A			

Review Questions

1. What are the most significant design considerations for computer-to-computer communication?

2. In the ISO-OSI paradigm, what are the main roles of the network layer? What distinguishes the network layer's packet delivery role from that of the data link layer?

3. In the OSI reference model, what is the objective of layer isolation?

4. Why is the OSI Reference Model so extensively used? What good did it do to establish itself as a data transmission standard?

5. Compare and contrast the OSI reference model with the TCP/IP model.

Further Readings

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